

Day 1 Mission: Investigate What Happens Inside a Computer

Today is not about learning computer components.

Today is about answering a question:

What actually happens inside a computer when I use it?

Most people use computers every day.

Very few can explain what the machine is doing.

Your mission is to investigate the system and discover the answer yourself.

Learning Rule

For every concept today:

Do NOT start with definitions.

Use this process:

Question ↓ Prediction ↓ Experiment ↓ Observation ↓ Explanation

Understanding comes from investigation.

Not memorization.

Investigation 1: What Is The Computer Doing Right Now?

Before opening anything, look at your computer.

Ask yourself:

What is the computer doing at this exact moment?

Questions:

- Is it completely idle?

- Are programs already running?
- What work is happening in the background?
- How can I observe it?

Your challenge:

Find a way to see what work the computer is currently performing.

Document everything you notice.

Investigation 2: Why Does A Computer Need A CPU?

Most people hear:

CPU executes instructions.

But that doesn't explain why it exists.

Start with the question:

What problem would exist if a computer had no CPU?

Write your prediction.

Then investigate.

Open:

- Chrome
- VS Code
- YouTube
- Spotify (or another application)

Observe:

- CPU usage
- Changes when applications start
- Changes when applications close

Questions:

- Which application requires more CPU?
- Why?
- What kind of work is the CPU performing?
- What happens when more work arrives?

Your goal:

Understand the role of the CPU through observation.

Not through definitions.

Investigation 3: Why Does A Computer Need RAM?

Question:

Why can't the computer run everything directly from storage?

Write your prediction first.

Then experiment.

Open applications one by one.

Observe:

- RAM usage
- Changes after opening applications
- Changes after closing applications

Next:

Open many browser tabs.

Observe:

- RAM growth
- System responsiveness
- CPU behavior

Questions:

- What changed?
- Why did it change?
- What information seems to be stored in RAM?

Your goal:

Discover the purpose of RAM.

Investigation 4: What Is Storage Actually Doing?

Question:

What is the difference between RAM and Storage?

Investigate.

Create:

- A text file
- An image
- A folder

Then:

- Restart your computer if possible
- Or close and reopen applications

Questions:

- What information remains?
- What information disappears?
- Why?

Document your observations.

Investigation 5: Why Does The Operating System Exist?

This is one of the most important questions today.

Imagine:

- No Windows
- No Linux
- No macOS

Question:

How would applications access hardware?

Write your prediction.

Then investigate.

Observe:

- Running applications
- Background services
- Processes
- System settings

Questions:

- Who decides which application runs?
- Who manages memory?
- Who manages files?
- Who manages devices?

Your goal:

Understand the purpose of the operating system.

Investigation 6: Cause And Effect

Now start creating workload.

Examples:

- Open many browser tabs
- Play a YouTube video
- Open VS Code
- Open additional applications

Observe:

- CPU changes
- RAM changes
- System responsiveness

For every action ask:

What changed?

Why did it change?

Which component was affected?

This is where real understanding starts.

Investigation 7: Failure And Limits

Every system has limits.

Try pushing your machine harder than usual.

Questions:

- What happens when RAM usage increases significantly?

- What happens when CPU usage becomes high?
- Does the computer behave differently?
- Which component becomes the bottleneck?

Observe carefully.

Do not rely on assumptions.

Final Reflection

Without searching online, answer:

1. Why does a CPU exist?
2. Why does RAM exist?
3. Why does storage exist?
4. Why does an operating system exist?
5. What surprised you today?
6. Which assumptions turned out to be wrong?
7. What happens when Chrome opens?

Use your own words.

Deliverables

1. Discovery Journal

Create:

day1-discoveries.md

For every investigation record:

- Prediction
 - Experiment
 - Observation
 - Conclusion
-

2. Evidence

Take screenshots showing:

- CPU observations
 - RAM observations
 - Running processes
 - Application behavior
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3. Explanation Recording

Record a 5-minute explanation:

"What actually happens inside a computer when I open an application?"

No notes.

No reading.

Only understanding.

Success Criteria

You succeed today if:

- CPU feels like a real system component, not a definition.
- RAM feels observable rather than theoretical.
- You understand the difference between RAM and storage.
- You understand why the operating system exists.
- You can explain what happens when an application starts.

If you only memorized definitions, repeat the investigations.

If you discovered why these components exist and observed them in action, you've completed the mission.